

SWETHA BAKKA — AI & DATA PLATFORM ENGINEER

Austin, TX | Open to Remote / Relocation

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PROFESSIONAL SUMMARY

AI & Data Platform Engineer with 10+ years of experience building scalable cloud systems, data pipelines, and enterprise platforms, now specializing in Generative AI and agentic workflows.

Experienced in designing LLM-powered applications using tools like Claude, Snowflake-integrated pipelines, and Python-based orchestration frameworks. Proven ability to integrate AI into data platforms for automation, anomaly detection, and decision intelligence.

Strong background in Snowflake, AWS, Terraform, and data governance, combined with a growing focus on AI agents, prompt engineering, and intelligent data systems.

Entrepreneurial builder with experience launching products end-to-end, blending technical depth with business impact.

AI & AGENTIC SYSTEMS EXPERIENCE

Built LLM-powered workflows using Claude and Python for data validation, anomaly detection, and automated insights generation

Designed prompt engineering strategies for structured data extraction, classification, and reasoning over enterprise datasets

Developed early-stage agentic workflows to automate data quality checks, lineage validation, and reporting

Integrated AI-driven logic into Snowflake-based pipelines using Python and SQL

Experimented with multi-step AI orchestration (Claude Code-style workflows) for task automation and decision support

Evaluated LLM performance across enterprise use cases including data governance, QA automation, and analytics

PROFESSIONAL EXPERIENCE

Data Quality & AI Consultant — AISOFT LLC (Sep 2024 – Present)

Designed and implemented AI-assisted data quality frameworks using Snowflake and Python, enabling automated anomaly detection and root cause analysis

Built LLM-driven validation workflows using Claude for data profiling, schema interpretation, and issue classification
Developed agent-like pipelines to automate lineage validation, reducing manual QA effort and improving traceability
Integrated Snowflake analytics with AI-generated insights for dashboards and operational decision-making
Partnered with DevOps teams to embed AI-driven validation into ETL pipelines, improving reliability and monitoring
Created data quality scorecards combining traditional metrics with AI-based insights

Founder & Creative Director — Loom & Luxe Expo / Swetha Reddy Designs (May 2024 – Present)

Founded and scaled a sustainable fashion brand with full ownership of product, platform, and go-to-market strategy
Built and launched Shopify-based e-commerce platform with integrated analytics
Applied data-driven decision making for pricing, demand forecasting, and inventory optimization
Led marketing campaigns leveraging analytics and audience segmentation
Managed vendor ecosystem, sourcing, and supply chain optimization

Software Engineer — Rocket Mortgage (Jul 2020 – Sep 2024)

Architected and deployed AWS infrastructure for enterprise data governance platforms using Terraform and IAM
Developed Collibra workflows for metadata management, issue tracking, and data stewardship
Automated CI/CD pipelines and QA validation frameworks using CircleCI and internal tooling
Led initiatives to improve data quality, lineage tracking, and compliance across enterprise systems
Built automation frameworks extendable to AI-driven classification and intelligent workflow orchestration
Partnered with legal, audit, and engineering teams to ensure regulatory compliance and system resilience
Mentored teams on SDLC best practices and evolving AI-assisted development workflows

Software Engineer — Rocket Central (Jan 2019 – Jul 2020)

Led implementation of data governance workflows across access management, data quality, and issue resolution
Designed reusable Terraform modules for AWS infrastructure provisioning (S3, Lambda, IAM)
Built QA automation frameworks for data ingestion pipelines using Java and Groovy
Collaborated with stakeholders to identify pipeline gaps and improve data reliability

QA Engineer / Analyst — Early Career (2010 – 2018)

Built and executed automation frameworks for enterprise applications and APIs
Developed regression testing suites and validation workflows for large-scale systems
Supported clinical and financial data pipelines ensuring SLA and data quality compliance
Created tools and automation scripts to improve QA efficiency and reporting

AI PROJECTS

AI Data Quality Agent: Built agentic workflow using Claude to detect anomalies and generate insights

Snowflake AI Integration: Designed pipelines combining SQL, Python, and LLM reasoning

LLM-Based Data Profiling Tool: Automated schema interpretation and rule generation

TECHNICAL SKILLS

AI/GenAI: Claude, Prompt Engineering, LLM Workflows, Agentic Systems

Data: Snowflake, SQL, Python, Data Governance

Cloud: AWS, Azure, GCP, Terraform, CI/CD

Programming: Python, Java, JavaScript

Tools: Colibra, JIRA, Postman

EDUCATION

M.S. in Engineering Management — Wayne State University

B.Tech in Electronics and Communications — JNTU Hyderabad